Guide for Building the Standard Schema and Keywords of Materials Research Data : NCMRD-Schema

2025.06.11.

Expert Committee for Materials Research Data Standardization

Contents

The rules for creating the schema

1. The NCMRD-Schema defines the standard schema of materials R&D data and and keywords used in the schema.
2. It follows the json-schema format (https://json-schema.org). json-schema file is the json file that defines the schema of the data.
3. Each keyword contains the following information.
   - description : description or definition of keyword
   - alias : vocabularies used with the same meaning as the keyword
   - type : the data type corresponding to the keyword

     Followings are examples of data types.
     ** string : String data*
     ** number : Numeric data*
     ** boolean : True or False*
   - unit : unit of the data corresponding to the keyword, mandatory for the data of type number

     The unit of all number data should be defined based on the "Appendix: List of Units Permitted"
   - examples : examples of the data corresponding to the keyword. must be useful when the data type is string

   (Important) The platform is built based on the information parsed from the json-schema file. Therefore, the spelling of these 5 terms must be correctly used in json-schema file.
4. To be integrated with international standard, all keywords are to be in English.
5. All keywords used in this schema must be written in lowercase, except for proper nouns.
   Ex) chemical information; Czochralski growth method

   Note) In the Korean Material Data System, internal variables are generated and used based on keywords. The internal variables use the exact words and spelling of the keywords, but each word's first letter is capitalized and the words are combined without spaces.
   ex) "structural property" → StructuralProperty
   "Czochralski growth method"→ CzochralskiGrowthMethod

6. All number type data can include both a value and an uncertainty, with the information of measurement. The uncertainty is provided as an absolute uncertainty, expressed in the same unit as the value.

   Ex)

```
▼ density : {
    description : mass per unit volum of a substance
    alias : value
    type : object
  ▼ properties : {
      ▶ value : { 3 props }
      ▶ uncertainty : { 1 prop }
      ▶ measurement : { 1 prop }
    }
    example : {"value":1.45, "uncertainty":0.03, "measurement":{...}}
  }
```

7. Multiple instances of “examples” are separated by semicolons (;). Multiple instances of “alias” are also separated by semicolons (;).

   Ex) annealed INCONEL625 plate; Al7075-T6; Al10Si5Mg; Fe; TiO2 on CNT

8. Units are displayed using superscripts only.

   ex)

V/m → V m^{-1}	m^2/s^2V → m^{2} s^{-2} V^{-1}
m/s → m s^{-1}	m^3/C → m^{3} C^{-1}
A/m^2 → A m^{-2}	$m^{1/3}/C$ → m^{1/3} C^{-1}
mm/year → mm year^{-1}	

9. Special symbols and formulas follow LaTeX notation.

   https://en.wikipedia.org/wiki/Help:Displaying_a_formula#Alphabets_and_typefaces

   ex)

\alpha	\mu	\epsilon	\theta	\lambda	\sigma	\cdot	\log	\ANGSTROM
α	μ	ϵ	θ	λ	σ	$\cdot$	$\log$	Å

(Appendix) List of Units Permitted

The list is based on the 7 SI base units, but the use of units widely used in the field of materials science is also permitted.

red = 7 SI base units

Time		Length		Mass	
year	year	m	meter	kg	kilogram
d	day	cm	centi meter	g	gram
h	hour	mm	milli meter	mg	mili gram
min	minute	{₩mu m}	micro meter	{₩mu g}	micro gram
s	second	nm	nano meter	ng	nanogram
ms	mili second	₩ANGSTROM	Angstrom	m_{e}	electron mass
{₩mu s}	micro second			Da	dalton
ns	nano second				
fs	femto second				

Electricity		Thermodynamic Temperature / Energy / Power		Amount	
A	Ampere	eV	electron volt	mol	mole
mA	milli Ampere	keV	kilo electron volt	mmol	mili mole
{₩mu A}	micro Ampere	meV	mili electron volt	{₩mu mol}	micro mole
nA	nano Ampere	J	Joule	nmol	nano mole
pA	pico Ampere	kJ	kilo Joule		
V	volt	pJ	pico Joule		
kV	kilo volt	W	watt		
mV	milivolt	K	kelvin degree		
{₩mu V}	micro volt	kW	kilo watt		
C	coulomb	MeV	mega electron volt		
{₩mu C}	micro Coulomb	mW	mili watt		
pC	pico Coulomb	{₩mu W}	micro watt		
e	charge of an electron				
S	conductance (Siemens)				
Ohm	resistance				
F	Faraday				

Luminous Intensity		Force / Pressure		Others	
cd	candela	N	Newton	at.%	atomic per cent
		{₩mu N}	micro Newton	wt.%	weight per cent
		kN	kilo Newton	ppm	part per million
		kgf	kilo gram force	%	per cent
		gf	gram force	HR	Hardness scale
		Pa	pascal	HB	
		MPa	mega pascal	HV	
		GPa	giga pascal	HK	
		Torr	Torr	Jones	specific detectivity
		bar	Bar	Hz	Hertz
		atm	atmosphere	kHz	kilo Hertz
				revolution	revolution, rotation
				degree	geometric angle
				L	liter
				mL	milli liter
				cycle	number of repeat
				T	tesla
				{₩mu L}	micro liter
				GHz	giga Hertz
				dB	decibel
				D	Debye